

EXPERIMENT NO. 6

Predictive Analytics using Linear Regression and

Model Performance Evaluation

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In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Step 1: Select and Load the dataset
housing = fetch_california_housing()

df = pd.DataFrame(
    housing.data,
    columns=housing.feature_names
)

df["MedHouseVal"] = housing.target

print("Step 1: Dataset Loaded Successfully")
print(df.head())

# Step 2: Import libraries and check dataset
print("\nStep 2: Dataset Information")
print("Shape:", df.shape)
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())

# Step 3: Data preprocessing and feature selection
X = df.drop("MedHouseVal", axis=1)
y = df["MedHouseVal"]

print("\nStep 3: Features and Target Selected")
print("Features:", X.columns.tolist())
print("Target: MedHouseVal")

# Step 4: Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

print("\nStep 4: Dataset Split")
print("Training Data:", X_train.shape)
```

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print("Testing Data:", X_test.shape)

# Step 5: Train Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)

print("\nStep 5: Linear Regression Model Trained Successfully")

# Step 6: Predict values using the trained model
y_pred = model.predict(X_test)

print("\nStep 6: Prediction Completed")
print("First 10 Predicted Values:")
print(y_pred[:10])

# Step 7: Visualize actual and predicted values
plt.figure(figsize=(8, 6))

plt.scatter(y_test, y_pred, alpha=0.5)

min_value = min(y_test.min(), y_pred.min())
max_value = max(y_test.max(), y_pred.max())

plt.plot(
    [min_value, max_value],
    [min_value, max_value],
    linestyle="--"
)

plt.xlabel("Actual House Values")
plt.ylabel("Predicted House Values")
plt.title("Actual vs Predicted House Values")
plt.show()

# Step 8: Evaluate the model
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("\nStep 8: Model Performance Evaluation")
print("Mean Absolute Error (MAE):", round(mae, 4))
print("Mean Squared Error (MSE):", round(mse, 4))
print("Root Mean Squared Error (RMSE):", round(rmse, 4))
print("R2 Score:", round(r2, 4))

# Step 9: Analyze regression coefficients
coefficients = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": model.coef_
})

print("\nStep 9: Regression Coefficients")
print(coefficients)
```

```
plt.figure(figsize=(10, 6))

plt.barh(
    coefficients["Feature"],
    coefficients["Coefficient"]
)

plt.xlabel("Coefficient Value")
plt.ylabel("Features")
plt.title("Regression Coefficients")
plt.show()

# Step 10: Interpret and display final result
comparison = pd.DataFrame({
    "Actual Value": y_test.values,
    "Predicted Value": y_pred
})

print("\nStep 10: Actual vs Predicted Values")
print(comparison.head(10))
```

Step 1: Dataset Loaded Successfully

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	\
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	

	Longitude	MedHouseVal
0	-122.23	4.526
1	-122.22	3.585
2	-122.24	3.521
3	-122.25	3.413
4	-122.25	3.422

Step 2: Dataset Information

Shape: (20640, 9)

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 20640 entries, 0 to 20639

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	MedInc	20640 non-null	float64
1	HouseAge	20640 non-null	float64
2	AveRooms	20640 non-null	float64
3	AveBedrms	20640 non-null	float64
4	Population	20640 non-null	float64
5	AveOccup	20640 non-null	float64
6	Latitude	20640 non-null	float64
7	Longitude	20640 non-null	float64
8	MedHouseVal	20640 non-null	float64

dtypes: float64(9)

memory usage: 1.4 MB

None

Missing Values:

MedInc	0
HouseAge	0
AveRooms	0
AveBedrms	0
Population	0
AveOccup	0
Latitude	0
Longitude	0
MedHouseVal	0

dtype: int64

Step 3: Features and Target Selected

Features: ['MedInc', 'HouseAge', 'AveRooms', 'AveBedrms', 'Population', 'AveOccup', 'Latitude', 'Longitude']

Target: MedHouseVal

Step 4: Dataset Split

Training Data: (16512, 8)

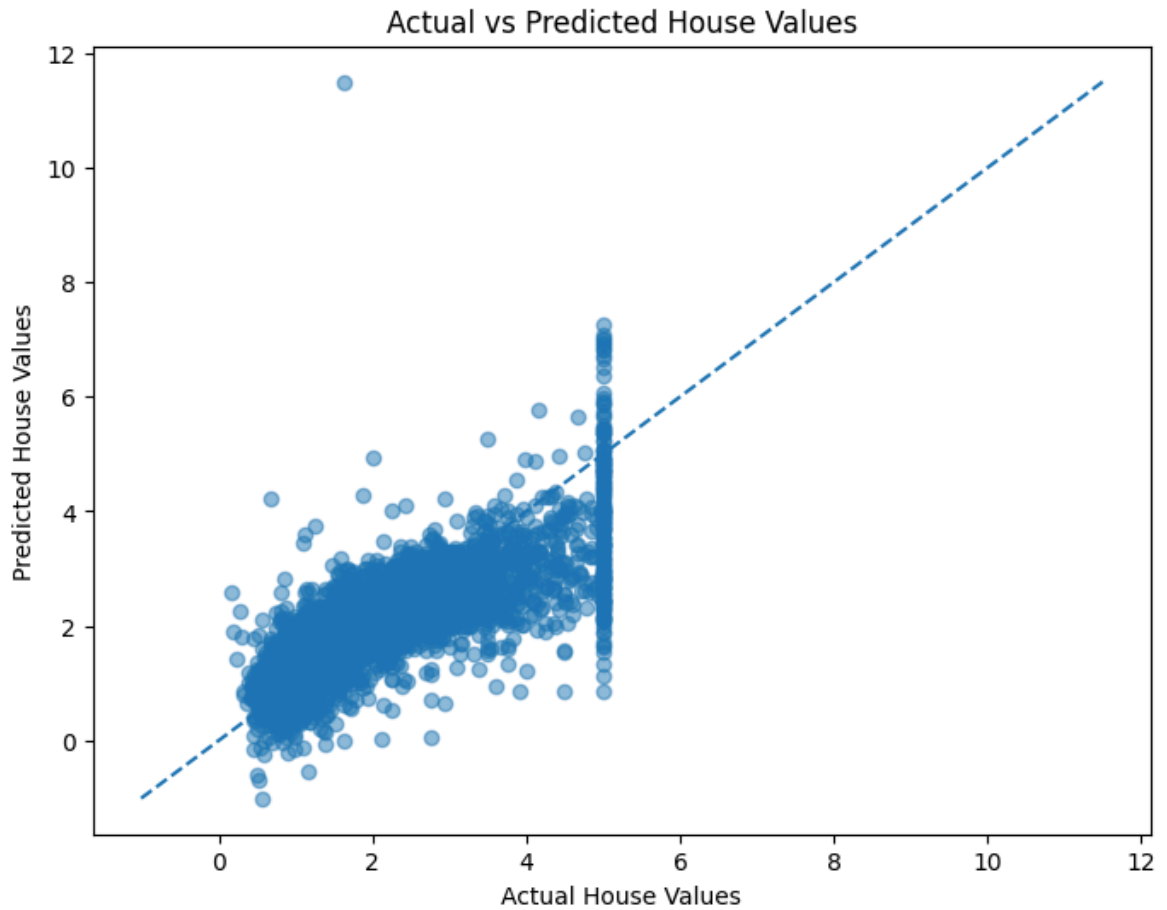
Testing Data: (4128, 8)

Step 5: Linear Regression Model Trained Successfully

Step 6: Prediction Completed

First 10 Predicted Values:

```
[0.71912284 1.76401657 2.70965883 2.83892593 2.60465725 2.01175367  
2.64550005 2.16875532 2.74074644 3.91561473]
```



Step 8: Model Performance Evaluation

Mean Absolute Error (MAE): 0.5332

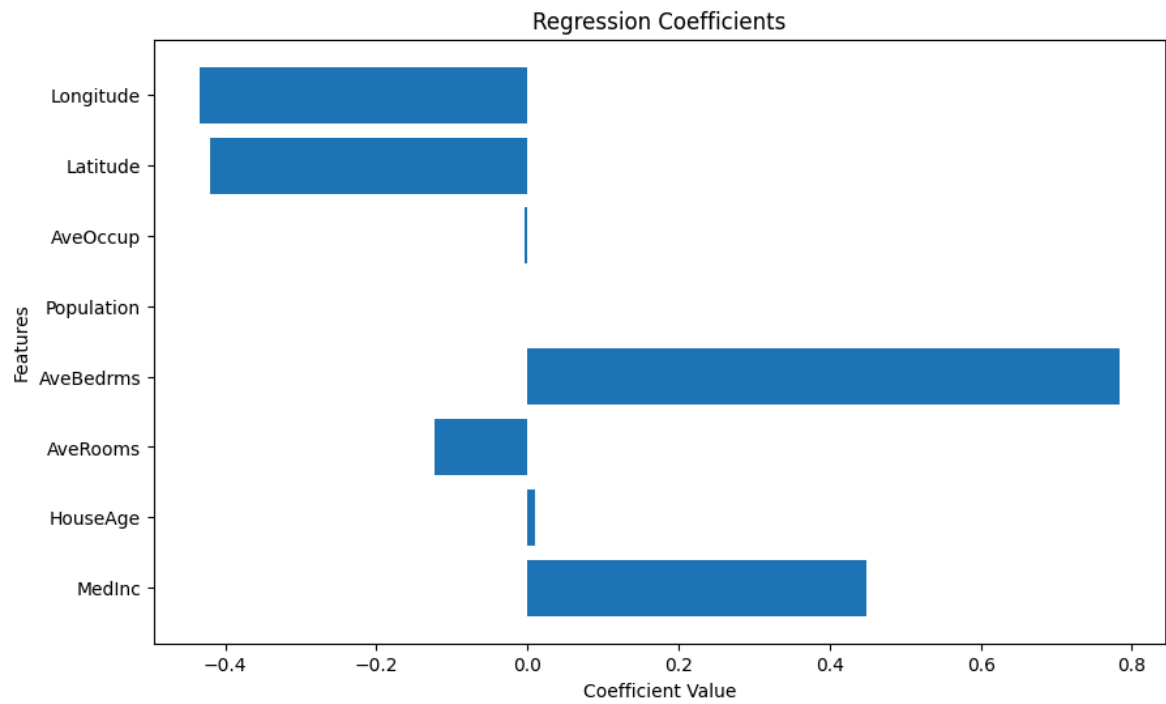
Mean Squared Error (MSE): 0.5559

Root Mean Squared Error (RMSE): 0.7456

R² Score: 0.5758

Step 9: Regression Coefficients

	Feature	Coefficient
0	MedInc	0.448675
1	HouseAge	0.009724
2	AveRooms	-0.123323
3	AveBedrms	0.783145
4	Population	-0.000002
5	AveOccup	-0.003526
6	Latitude	-0.419792
7	Longitude	-0.433708

**Step 10: Actual vs Predicted Values**

	Actual Value	Predicted Value
0	0.47700	0.719123
1	0.45800	1.764017
2	5.00001	2.709659
3	2.18600	2.838926
4	2.78000	2.604657
5	1.58700	2.011754
6	1.98200	2.645500
7	1.57500	2.168755
8	3.40000	2.740746
9	4.46600	3.915615